

Automated Customer Review Analysis

NLP Pipeline for Product Recommendations &
Insights

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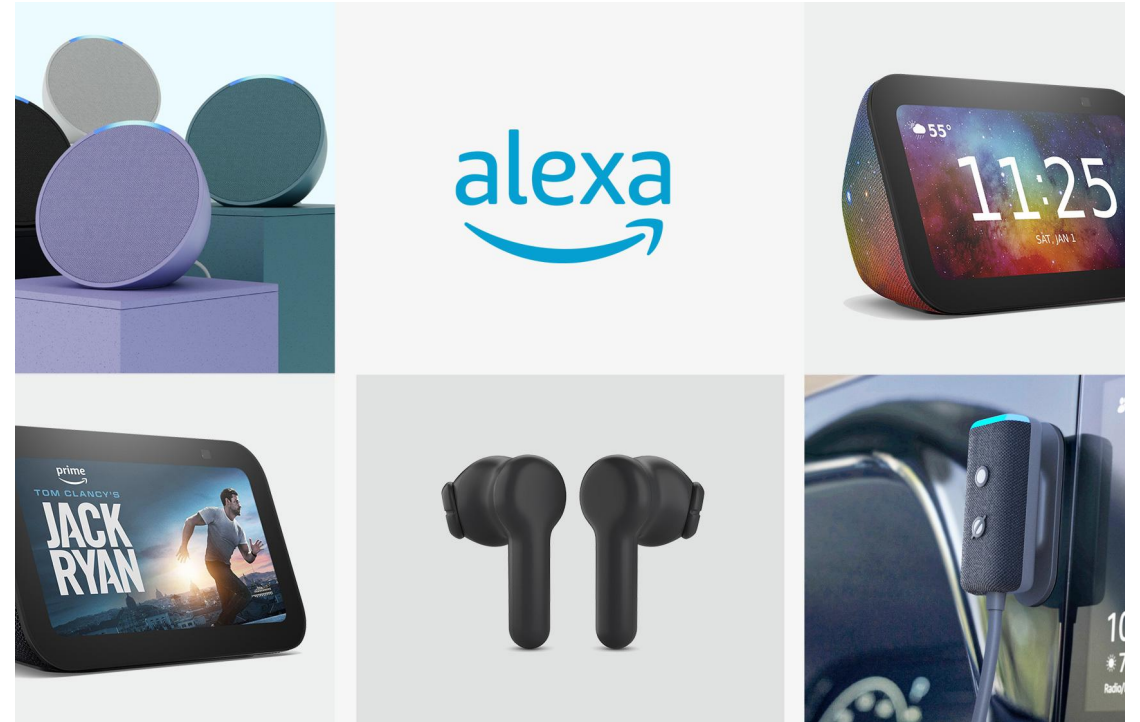
IronHack AI Engineering · May 2026

The Business Challenge

Why Automate Analysis?

- **Unscalable Volume:** Millions of daily reviews are impossible to process manually.
- **Market Intelligence:** Immediate feedback on quality, defects, and buyer profiles.
- **Actionable Data:** Transforming raw text into structured marketing materials.

"Goal: Build an end-to-end pipeline that transforms reviews into publish-ready recommendation articles."



System Architecture



01. Preprocessing

Cleaning, deduplication, and sentiment mapping from star ratings.



02. Classification

RoBERTa fine-tuned for high-precision sentiment detection.



03. Clustering

TF-IDF + PCA + K-Means for thematic product grouping.



04. Generative AI

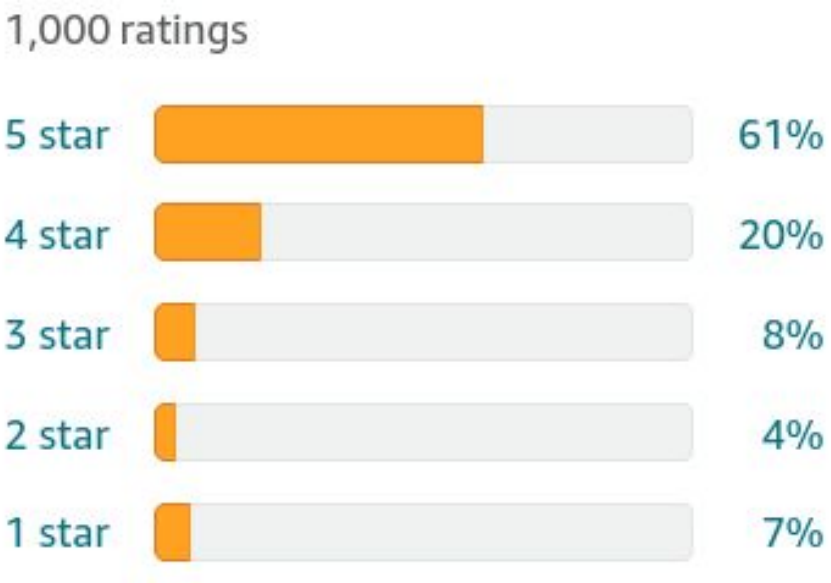
GPT-4o-mini structured summarization and recommendation generation.

Dataset Profile

Metric	Value
Total Clean Reviews	4,805
Unique Products	23
Top Categories	Electronics / Office
Average Rating	4.64 / 5.0

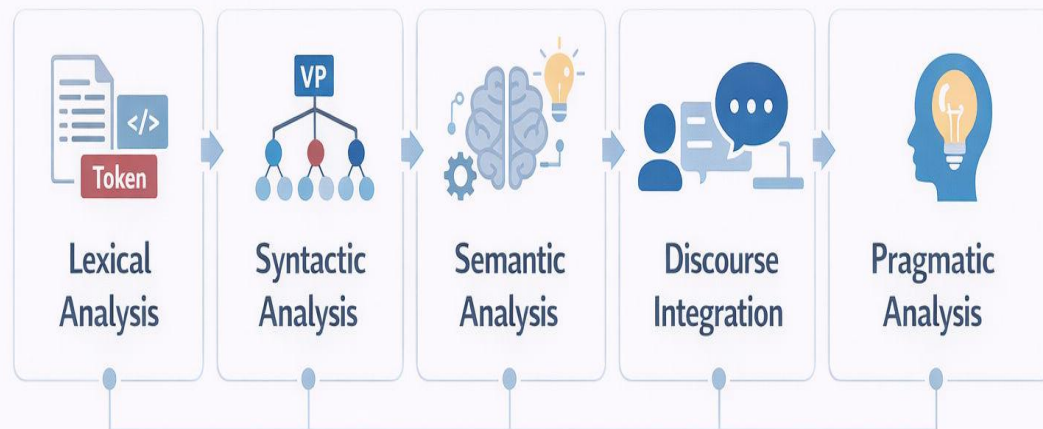
Preprocessing Pipeline:

- 1. Text normalization (lowercase/strip)
- 2. Sentiment mapping (1-2★ Neg, 3★ Neu, 4-5★ Pos)
- 3. Exact text deduplication



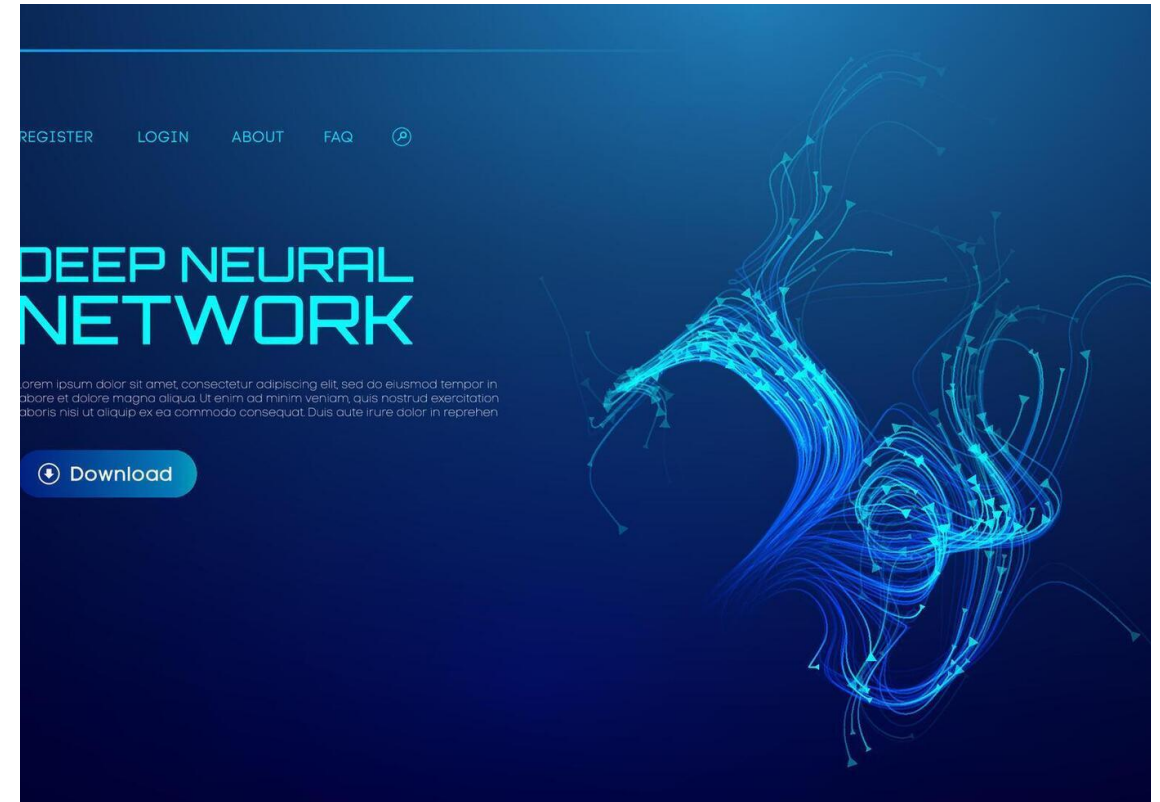
Exploratory Data Analysis

What Are the 5 Stages of NLP?



Class Imbalance

93.8% Positive reviews require class-weighted learning



Review Lengths

Median 21 words. Justifies a 128-token context window for

Transformers.

Sentiment Classification Benchmarks



Fine-tuning RoBERTa provided a **+19 point F1 improvement** over traditional baselines.

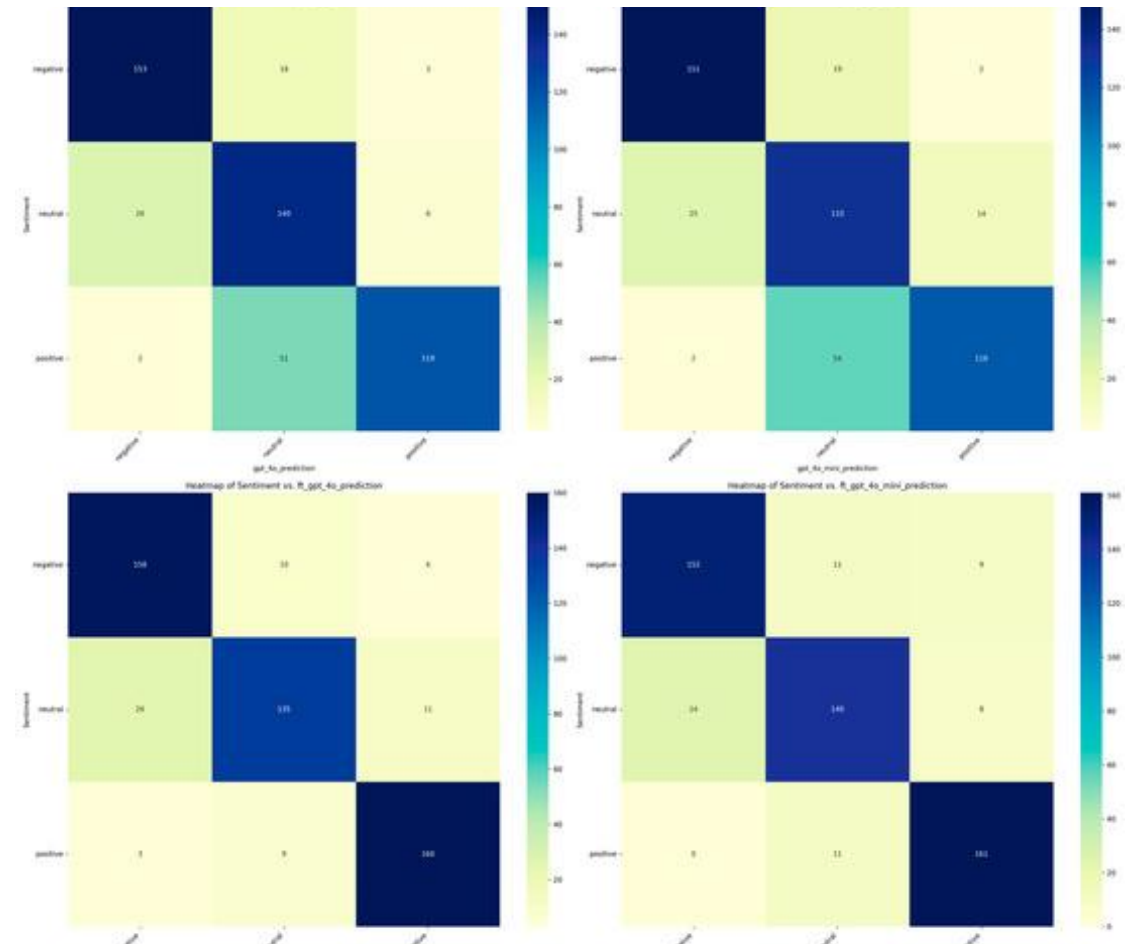
Fine-Tuning: Class-Weighted Loss

The Solution

Standard training ignores the 4% neutral class. We implemented **CrossEntropyLoss** with class weights to penalize minority errors more severely.

Impact

Increased Neutral/Negative F1 scores significantly.
The model captures "hidden" complaints better than zero-shot alternatives.

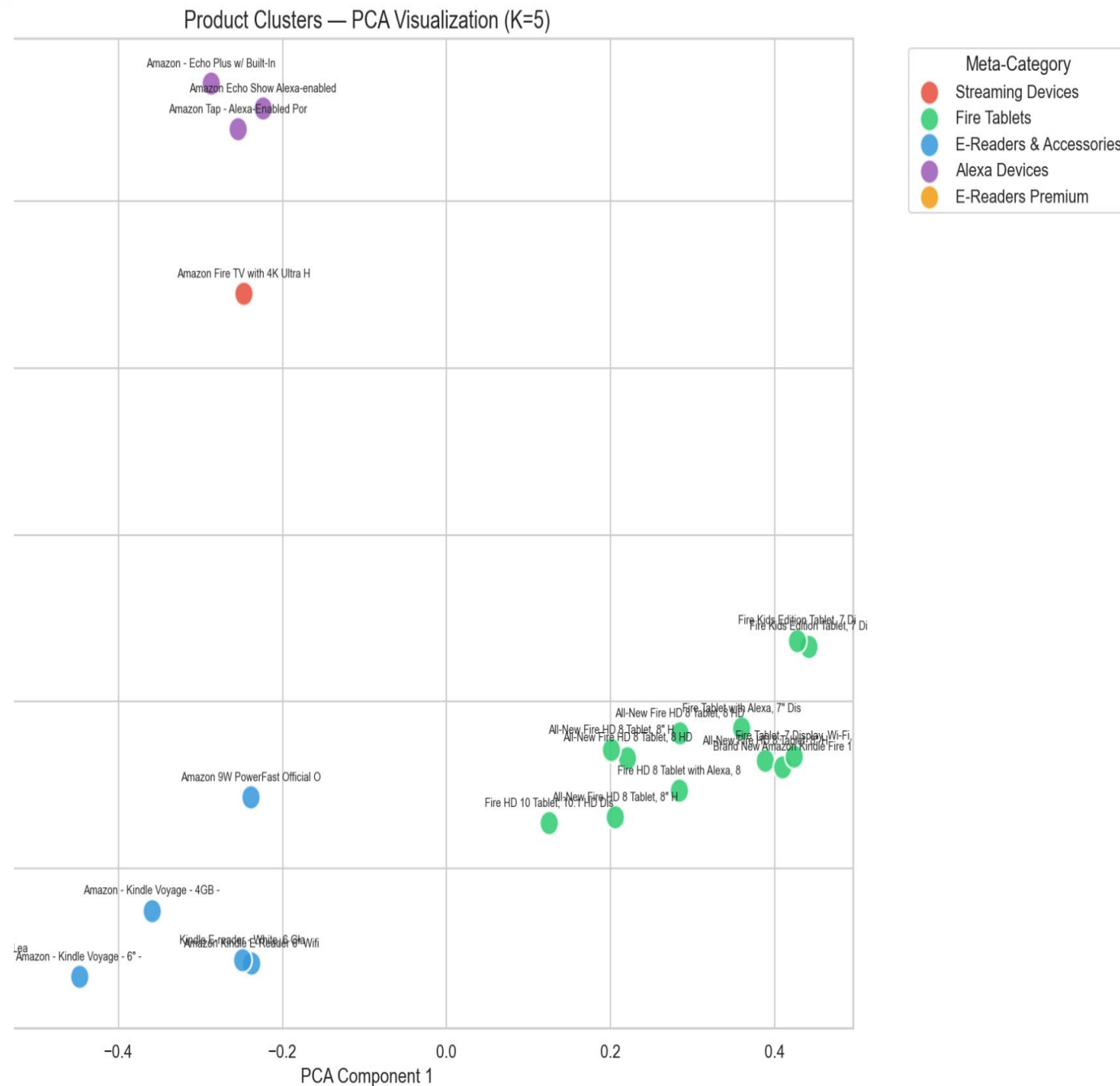


Product Clustering

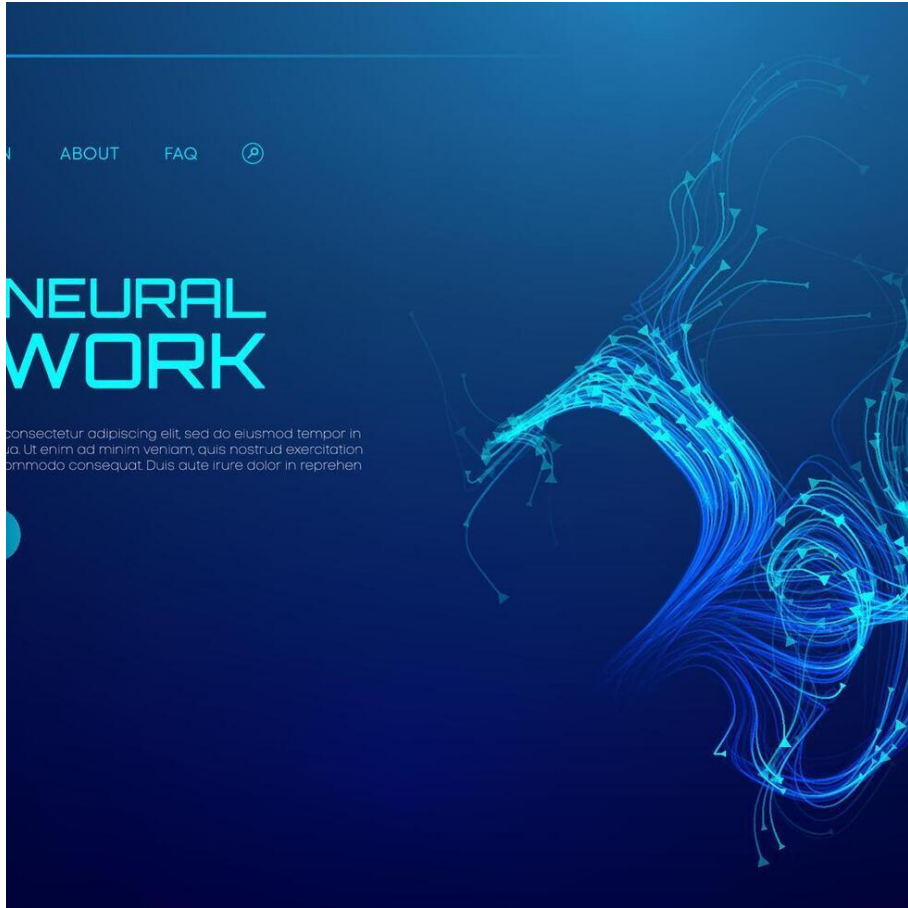
Unsupervised Thematic Discovery

Using **TF-IDF** and **PCA** (50 components), we grouped 23 products into 5 semantically coherent clusters.

- **K-Means:** Chosen for linear scalability and interpretability.
- **Optimal K:** Determined via Silhouette Score (0.40).
- **Result:** Discovered Amazon's actual product lines (Alexa, Kindle, Fire) with zero human labeling.



AI Content Generation



GPT-4o-mini Summarization

Structured prompts transform hundreds of classified reviews into professional product guides.

- 1. Executive Summary**
- 2. Key Strengths** (from Pos reviews)
- 3. Areas for Improvement** (from Neg/Neu)
- 4. Ideal Buyer Profile**

Actionable Insights

*"The Echo Plus is celebrated for its seamless integration with smart home devices [...] ideal for users looking to create an extensive ecosystem, while the Fire Tablet remains the top choice for budget-conscious families looking for durability." —
— Automated AI Product Guide (Alexa Devices Category)*

**Automated AI Product Guide (Alexa Devices
Category)**

This final deliverable is publish-ready for e-commerce blogs or internal marketing reports.

System Deployment

84%

Classification
Accuracy

10/10

Code Quality (Pylint)

23

Products Analyzed

3

Live Cloud Apps



Technical Stack

Core ML

HuggingFace Transformers, PyTorch, Scikit-learn, Pandas.

Deployment

Streamlit Cloud, HuggingFace Hub, Gradio Spaces.

MLOps

python-dotenv (security), Pylint (standards), Google-style docstrings.

Scalability: The "Upload & Analyse" feature allows the pipeline to process any CSV of reviews, making it sector-agnostic.

Questions?

Thank you for your
time.



/franromlob



Live Demo App

Links

Streamlit Community Cloud:

<https://projectautomatedcustomerreviews-yjfrjpfxn2s3fnyqdqgrlk.streamlit.app/>

HuggingFace Hub: <https://huggingface.co/FranMRL/roberta-amazon-sentiment>

HuggingFace Spaces + Gradio: <https://huggingface.co/spaces/FranMRL/amazon-review-sentiment>